Titanic Survival Prediction

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1. Introduction

For this project I decided to choose the Titanic dataset from Kaggle, which includes 891 rows of passenger information containing the respective ages, sex, class, and fare of each passenger. The goal was to predict who survived and did not. I was drawn to this dataset the most because it incorporates machine learning with a well known historical disaster, making the process a bit more tangible and engaging.

2. Description of the dataset

As stated earlier, the dataset contains 891 rows and 12 columns of passenger information. The features consist of multiple columns such as age, sex, and ticket class, while the target label is a single column

Indicating whether the passenger survived(1) or did not(0). The dataset contained some missing values, which were handled in 3 different ways. The cabin column was dropped entirely because 77% of it was missing, the age column had 177 missing values which were filled in using the median age, and the 2 missing embarked values were filled using the most common port. Additionally, the target classes are imbalanced, 549 passengers did not survive while 342 did, giving us a survival rate of only 38.14%. Due to this imbalance, The F1 score will be used as the main evaluation metric instead of accuracy.

3. Models used

Three machine learning models were trained and compared in this project

- Logistic Regression

  Logistic Regression achieved an accuracy of 0.804 and an f1 score of 0.724, clearly dividing the survivors from non survivors, which works reasonably well, but misses some of the more complex patterns in our data.

- Decision tree

  Decision tree achieved an accuracy of 0.821 and an f1 score of 0.758, making it the best performing model in the first step. It works by asking a series of yes or no questions about each passengers features until it reaches a prediction

- Random Forest

  Random forest achieved an accuracy of 0.816 and an f1 score of 0.752, by training many decision trees and combines their answers, which usually reduces overfitting compared to using a single tree, and in this case scored lower than the single decision tree

I selected the decision tree as the best model based on its highest f1 score of 0 .758.

4. Neural network

I built the neural network using keras with two dense layers of 64 and 32 neurons using ReLU activation,

A dropout layer of 0.3 between them to reduce overfitting and a sigmoid output layer. It was trained on 50 epochs with a 20% validation split. loss curve showed that the training loss started high but dropped sharply in the first 10 epochs before flattening out, with the validation loss being similar, meaning the model was actually learning and not just overfitting. The accuracy curve showed both training and validation accuracy rising gradually and staying close throughout training, leaving us with scores of .750 for training accuracy and .727 validation accuracy.

5. Results and comparison

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.804	0.793	0.667	0.724
Decision Tree	0.821	0.794	0.725	0.758
Random Forest	0.816	0.781	0.725	0.752
Neural Network	0.709	0.631	0.594	0.612

The neural network was the weakest model with an f1 score of .612, far below the decision trees .758.

This is probably because the titanic dataset is small, only containing 891 rows, and neural networks usually need much larger datasets to outperform simpler models. The decision tree remains the best overall model for this task outmatching every other model.

6. What I Learned

What I learned from this project is that using a more advanced model doesn't always mean getting better results. The neural network was the most complex technique I used, but it still scored lower than all three models, even though it was more complex. This taught me that the size of the dataset matters a lot. The Titanic dataset has only 891 rows, and neural networks need much more data than that to perform well.

I also learned you have to prepare the data correctly before training. You can not just rush in, simple steps like filling in the missing age values and converting the sex and embarked columns into numbers had an obvious impact on how well each model performed.